REALISING A GENERIC POINT OF A FIBRE COMPONENT OVER A VALUATION RING BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5093-5125 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5127-5164, the complete author proof (36 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The author's statement contains a reference environment citing EGA, so the card carries a bibliography fragment. That fragment reproduces, field for field and without editing, the @ARTICLE{EGA} entry from the Stacks Project snapshot bibliography shipped with this corpus at focused-seeds/AGP/sources/stacks-snapshot-20261011/my.bib (same upstream Stacks Project snapshot). No entry text was altered and no citation was reconstructed from memory.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Spectra of valuation rings, closures viewed as integral subschemes, and generic points of components of a fibre; the construction is retained.

Source reference. Original lemma 01J8: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5164–5164. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *This can be found in the proof of* [1, IV Proposition 12.1.1.5] *Let $f : X \to S$ be a morphism of schemes. Let $y \in X$ be a point with image $t \in S$. Denote $Y \subset X$ the closure of $\{y\}$ viewed as an integral closed subscheme of X. Let $s \in S$ and let $x \in Y_s$ be a generic point of an irreducible component of Y_s. There exists a cartesian diagram*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

with the following properties:

(1) *S' is the spectrum of a valuation ring with generic point t' and closed point s',*
(2) *$g(t') = t$ and $g(s') = s$,*

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(3) *there exists a point* $y' \in X'_{t'}$ *which is a generic point of an irreducible component of* $(S' \times_S Y)_{t'} = Y_t \times_t t'$ *and satisfies* $g'(y') = y$,
(4) *denoting* $Y' \subset X'$ *the closure of* $\{y'\}$ *viewed as an integral closed subscheme of* X' *there exists a point* $x' \in Y'_{s'}$ *which is a generic point of an irreducible component of* $Y'_{s'}$ *with* $g'(x') = x$.

Proof. We choose a valuation ring R, we set $S' = \mathrm{Spec}(R)$ with generic point t' and closed point s', and we choose a morphism $h : S' \to X$ with $h(t') = y$ and $h(s') = x$. See Schemes, Lemma 01J8. Set $g = f \circ h$ so that $g(t') = t$ and $g(s') = s$. Consider the base change

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle\sigma}\uparrow\downarrow & & \downarrow \\ S' & \xrightarrow{g} & S \end{array}$$

We obtain a section σ of the base change such that $h = g' \circ \sigma$.

Of course σ factors through the base change $S' \times_S Y$ of Y as h factors through Y. Let $y' \in X'_{t'} \subset X'$ be the generic point of an irreducible component of the fibre

$$(S' \times_S Y)_{t'} = Y_t \times_t t'$$

containing the point $\sigma(t')$, i.e., such that $y' \rightsquigarrow \sigma(t')$. Since $g'(y') \in Y_t$ and $g(y') \rightsquigarrow g(\sigma(t')) = y$ we find that $g'(y') = y$ because y is the generic point of the fibre Y_t. Denote $Y' \subset X'$ the closure of $\{y'\}$ in X' viewed as an integral closed subscheme. Then σ factors through Y' as $\sigma(t') \in Y'$. Choose a generic point $x' \in Y'_{s'}$ of an irreducible component of $Y'_{s'}$ which contains $\sigma(s')$, i.e., we get $x' \rightsquigarrow \sigma(s')$ and hence $g'(x') \rightsquigarrow g'(\sigma(s')) = x$. Again as x is a generic point of an irreducible component of Y_s by assumption and as $g'(Y') \subset Y$ we conclude that $g'(x') = x$. □

References

[1] Dieudonné, Jean; Grothendieck, Alexander. *Éléments de géométrie algébrique*. Inst. Hautes Études Sci. Publ. Math., volumes 4, 8, 11, 17, 20, 24, 28, 32, 1961–1967.